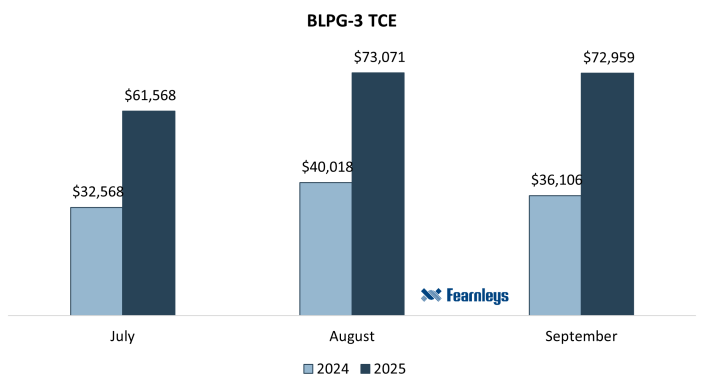
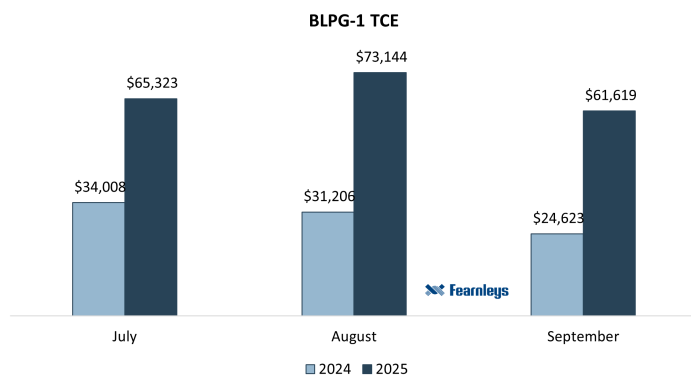
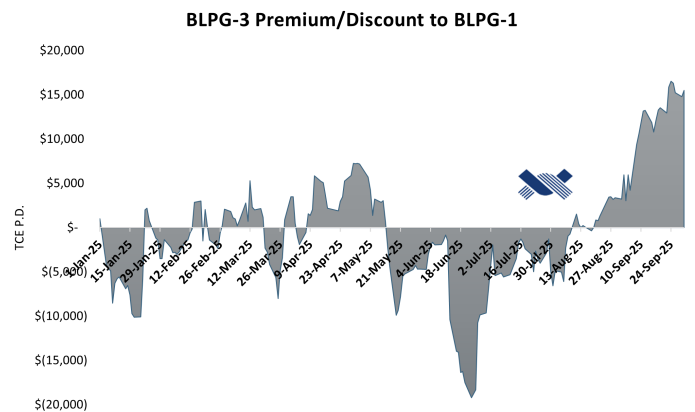
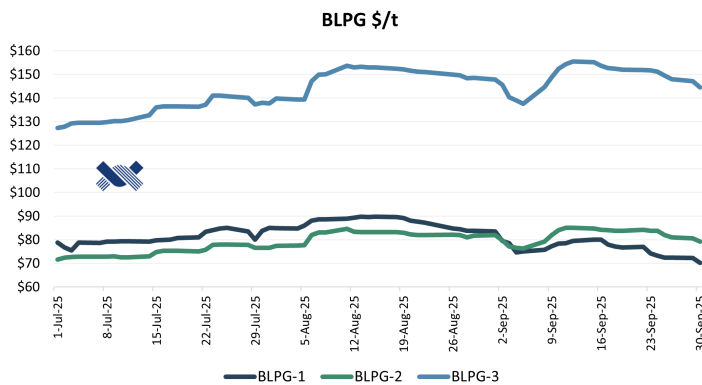


VLGC Quarterly Report - Q3 2025

Oct 9, 2025

Mia Willoch

Freight rates & Earnings



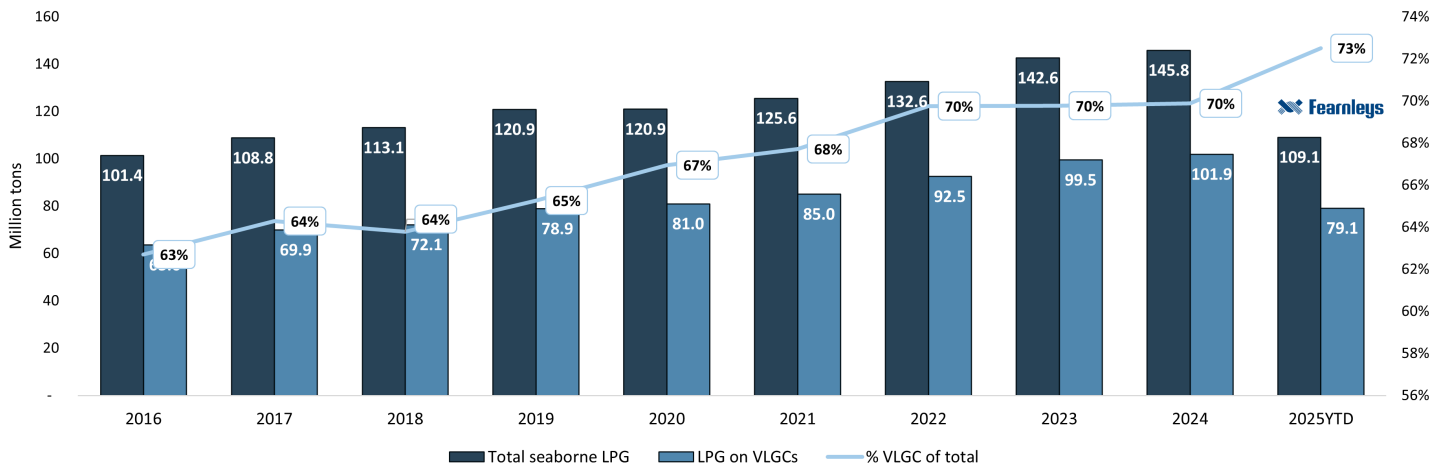
The average daily spot earnings across the three index routes reached \$75,000 per day in Q3, driven by a tighter market environment. This elevated the year-to-date average to \$54,000 per day, marking a \$10,000 increase from the post-Q2 figure—a 4.8% quarterly rise and a 45.6% year-on-year gain.

Q3 was largely defined by efforts to establish a new "normal" following the significant disruptions triggered by "Liberation Day" in April. The quarter was notably impacted by the widening spread between West and East cargo premiums, *as can be seen on above graph*. This shift was primarily driven by uncertainty surrounding the USTR and its pending rulings, which created confusion and concern as the fixing window approached the October 14th implementation date.

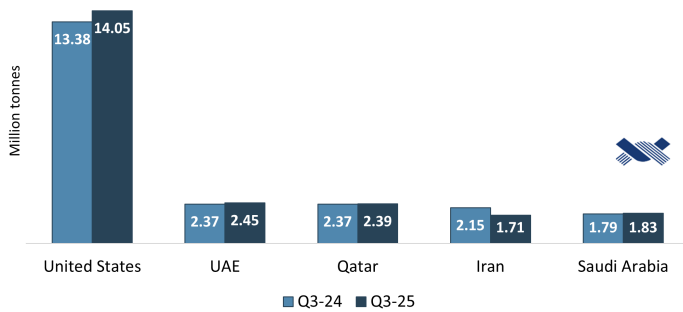
At the time of writing, operational clarity remains elusive, though consensus suggests the system will take effect next week as scheduled. The U.S. Customs and Border Protection has posted payment details for the fees, and they emphasize that it is entirely up to the Operator to determine whether the ship is due to pay. The fees are intended to target U.S. imports—specifically laden arrivals, which do not typically include VLGCs. However, ships that are Chinese-owned and/or operated will be subject to the fees regardless of arrival mode or country of build.

This regulatory uncertainty had a tangible impact on freight rates in September. Affected ships began repositioning toward the Middle East, and a notable premium emerged for West cargoes. At its peak, the premium reached \$16,500 per day, with the monthly average settling at \$11,300 per day. Nevertheless, if the premium becomes sufficiently attractive, even affected ships may resume lifting in the U.S., which could help rebalance the market and restore equilibrium.

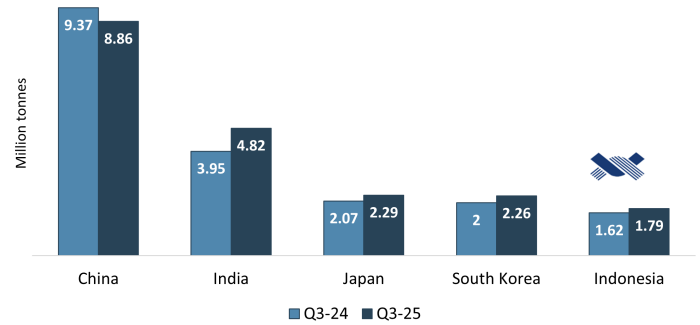
Exports and Imports



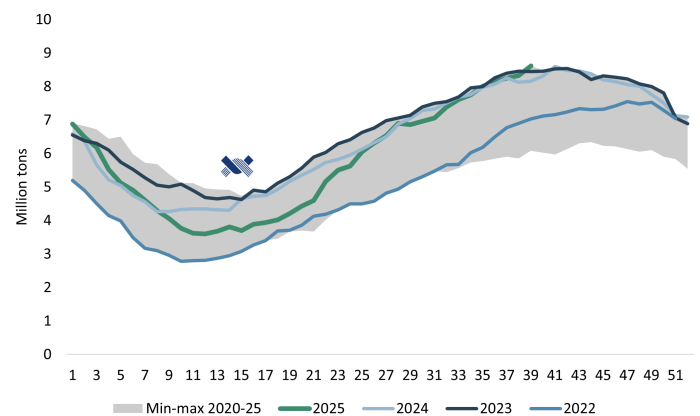
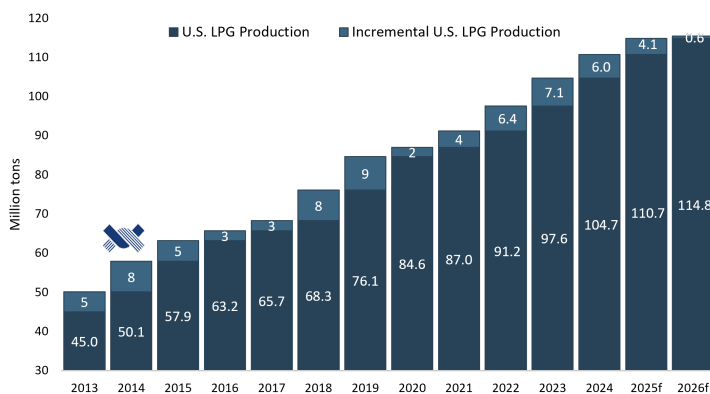
Top Exporters



Top Importers



U.S. Production and Terminal Capacity Expansion



U.S. LPG production continues to rise, with total output increasing by 75% over the past ten years. While growth is expected to moderate to approximately 5% over the next three years, export capacity is projected to expand significantly during the same period. Domestic consumption remains stable, as do export levels, which consistently hover around 60%. All residual volumes of LPG that is not domestically consumed, will be priced to clear out - enabling flexible exports. Furthermore, production from the Permian Basin is projected to grow at an annual rate of 7% through 2030, outpacing crude oil growth by 4%.

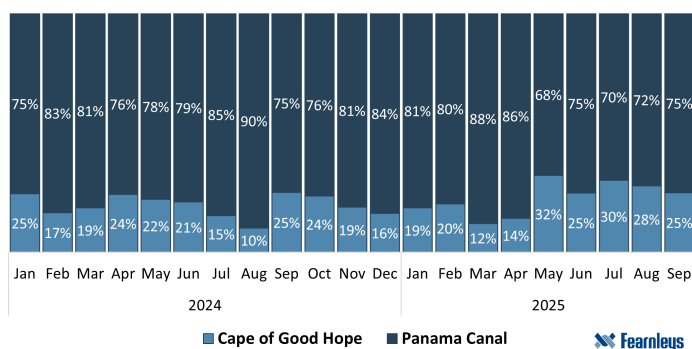
Propane inventories in the U.S. are now at the highest point its been in the past five years, with 2015 being the previous year recording these levels of builds. This could give guidance that a price correction is ahead, resulting in increased

exports available.

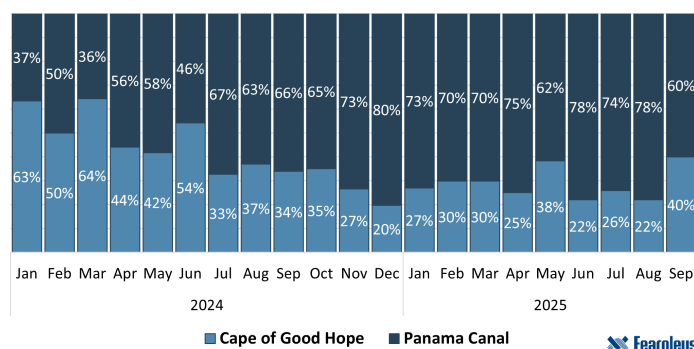
VLGCs account for 82% of all seaborne LPG exports from the U.S., totaling 52 million tons in 2024 and 41.25 million tons year-to-date in 2025—a 6.4% increase year-on-year. The majority of the volumes go to the Far East, though 15-20% head to Europe & Africa, which is a substantially shorter voyage.

The most influential factor in fleet utilization is the sailing patterns chosen for both laden and ballast voyages. Most laden ships heading to the Far East from the U.S. typically transit via the Panama Canal, which offers a significantly shorter route compared to the alternative around the Cape of Good Hope. Activity in the Suez Canal remains very limited, with very little VLGC traffic recorded. Year-to-date, 77% of laden voyages have passed through the Panama Canal, down 3% from the 2024 average of 80%. Conversely, 23% of ballasting ships returning to the U.S. from the East have opted for the Cape of Good Hope route—up from 20% in 2024, reflecting a 3% increase.

Laden VLGCs from USGC/USEC to Far East



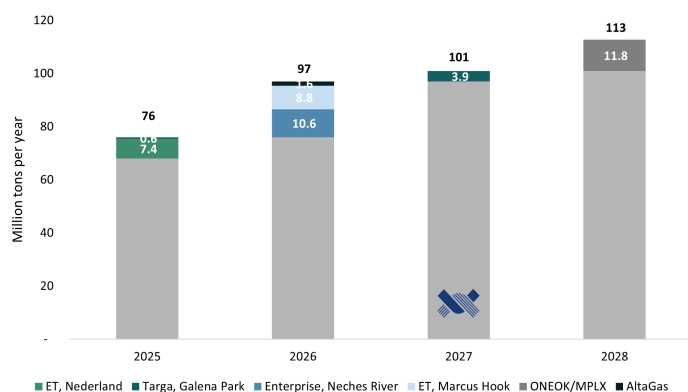
Ballast VLGCs from Far East to USGC/USEC



The Panama Canal remains a critical bottleneck in the VLGC market due to its unpredictable operational capacity and volatility. In Q3, congestion levels typically seen during peak winter months emerged unexpectedly, catching the market off guard. Auction fees surged to nearly \$1 million, significantly tightening capacity. As a result, the share of ballasting ships returning to the U.S. via the Cape of Good Hope (COGH) rose from 22% in August to 40% in September.

This congestion was primarily driven by above-average ship arrivals, and since VLGCs hold lower transit priority, they are particularly vulnerable to delays. In 2025, the Long-Term Slot Allocation (LoTSA) system has been fully operational, meaning a substantial portion of transit slots in the Neo-locks were auctioned off the year prior. This has led to full canal utilization, leaving limited room for non-booked ships. Additionally, high demand for Neo-slots is consuming both time and navigational channel capacity, reducing availability at the Panamax locks. The full utilization of Neo-locks has created a leapfrog effect, forcing non-booked ships to idle while waiting for transit opportunities. The Panama Canal Authority (PCA) has announced the continuation of the LoTSA system into 2026, albeit with some modifications.

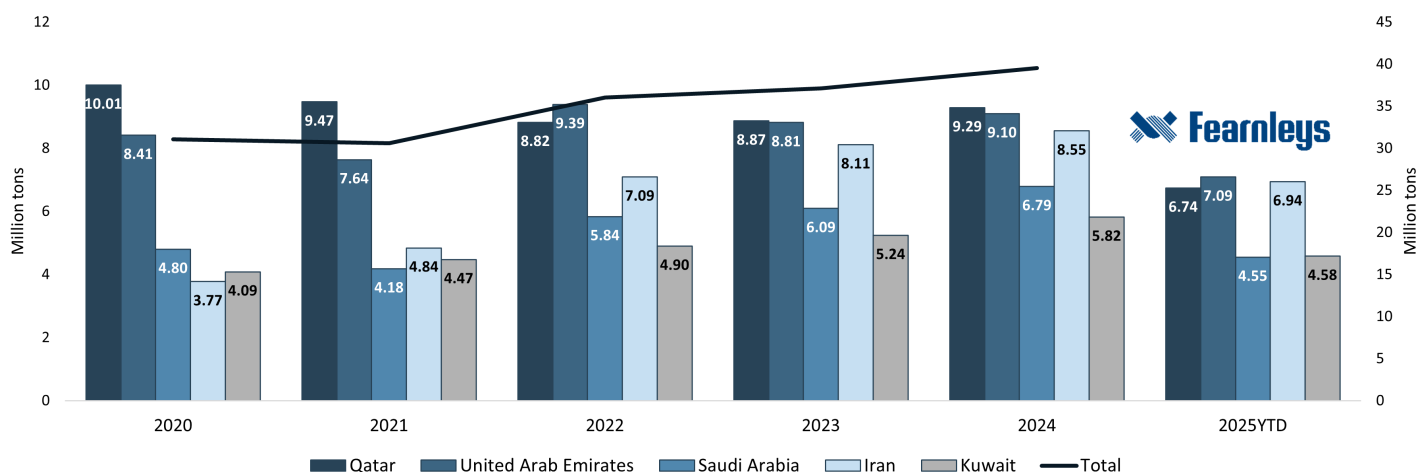
Another factor contributing to canal congestion is the emergence of VLECs transporting ethane from the U.S. to China. These voyages exclusively transit via the Panama Canal and do not use the COGH route. In 2025, the average number of VLEC transits has reached nine ships per month. With continued growth in U.S. exports of both LPG and ethane, this trend is expected to intensify.



Total U.S. export capacity is set to increase from 68 to 113 MTPA by 2028 with the completion of these projects. This would boost export capacity by 66% from 2024 to 2028—a substantial jump that is warmly welcomed by a growing fleet. What will materialize as the *actual* incremental exports is yet to be seen; however, it will likely be less than the full capacity numbers. Nevertheless, this move by U.S. energy suppliers further underscores the market's desire to expand and grow.

It should be noted that both the Nederland and Neches River facilities will offer flexible capacity for LPG and ethane. Given the recent growth in U.S. ethane exports as well as the VLEC orderbook (170% to fleet), it is expected that a portion of these volumes will be allocated to ethane rather than LPG.

The Middle East



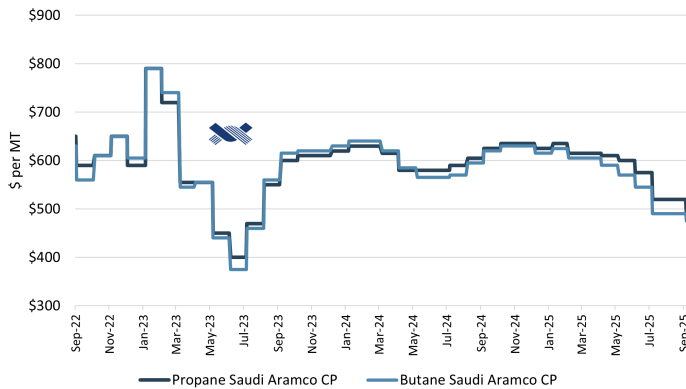
The ongoing trade tensions between the U.S. and China, has caused disruptions in cargo flows, solidifying LPG exports from the Middle East, which have surpassed 30 million tons year-to-date. Key contributors include Qatar, the UAE, Saudi Arabia, and Iran. Notably, 90% of Iranian LPG exports are destined for China, with 96% of these volumes transported by VLGCs, primarily by the dedicated fleet segment.

New natural gas developments in Qatar, including the North Field projects, are expected to come online in 2026 and 2027, bringing incremental LPG volumes. An accumulated 6.2 million tons per annum (MTPA) is anticipated, which would raise the region's export capacity from 46 to 52 MTPA.

In parallel, the reversal of OPEC+'s two-year production cut is projected to increase output, potentially adding the equivalent of 4–5 additional cargoes per month from the Middle East. If these volumes translate directly into VLGC loadings, it would represent a 5% increase in exports. Looking even further ahead, Saudi Arabia plans to boost natural gas production by 60% between 2021 and 2030, primarily from the Jafurah Field. Year-to-date, Saudi Arabia accounts for 7% of all VLGC liftings. The anticipation of future incremental volumes is 5-10 MTPA.

In July, the U.S. and Indonesia signed a trade agreement under which Indonesia will increase its purchases of U.S. petroleum products by \$13 billion, in exchange for reduced tariffs on Indonesian exports to the U.S. Indonesia currently sources 60% of its LPG imports from the U.S., and under the new agreement, it will shift its remaining demand—equiv-

alent to 2–3 MTPA—away from the Middle East. This shift could increase Middle East tonne-miles, as displaced volumes may now be redirected to the Far East if not India, depending on trade dynamics.



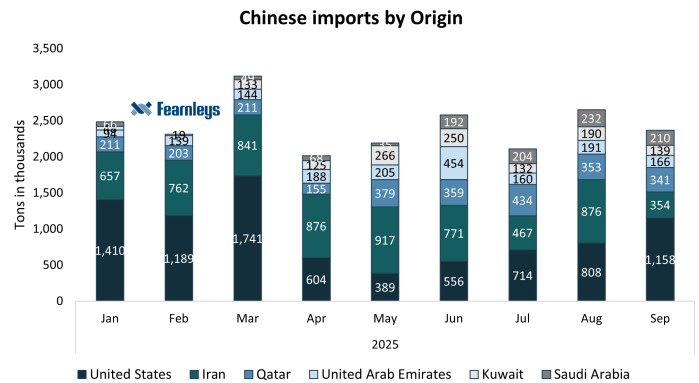
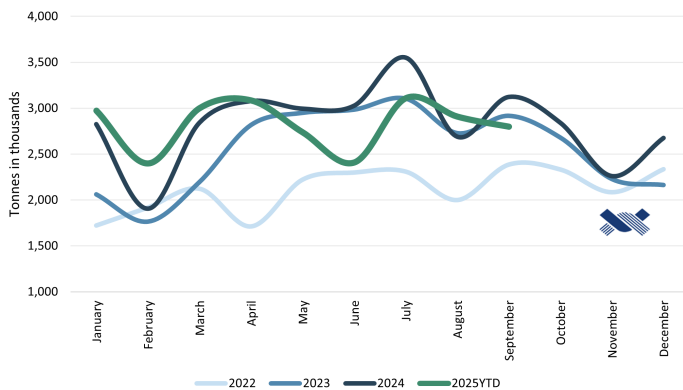
The LPG product market took an unexpected turn as Saudi Aramco lowered its October Contract Prices (CPs) to \$495/mt for propane (down 3.1%) and \$475/mt for butane (down 4.8%). These levels came in well below market expectations, which had anticipated propane CPs above \$540/mt.

This sharp adjustment appears to be a strategic move aimed at stimulating term buying interest and enhancing Saudi competitiveness against rising U.S. exports, which are being supported by ongoing terminal expansions.

The price cut could unlock new arbitrage opportunities and boost demand from Asian buyers, particularly in price-sensitive markets. There is therefore potential for increased liftings from Ras Tanura and Yanbu.

In parallel, if Asian buyers shift preference toward Saudi-origin cargoes, this could displace long-haul U.S. barrels heading East, potentially reducing overall tonne-mile demand and reshaping trade flows.

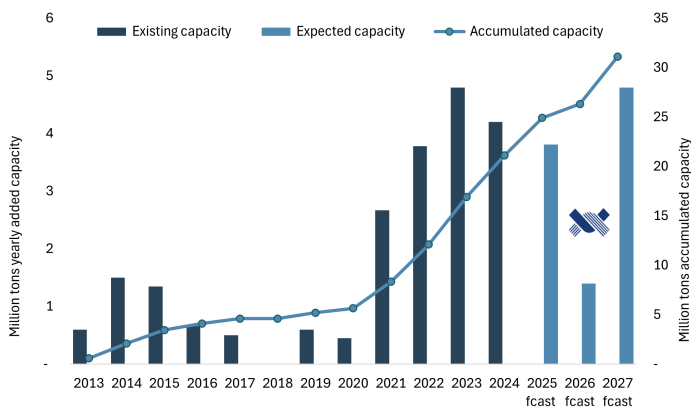
Chinese Demand



Following “Liberation Day” in April, cargo flows have been reshuffled, underscoring the high flexibility of the LPG market—though this adaptability comes at the cost of increased logistics expenses. Compared to the same period last year, Chinese VLGC imports remain steady at 25.95 million tons.

The most notable trade shifts include increased U.S. LPG imports into Japan and South Korea, while Chinese imports from the U.S. declined sharply. In response, the UAE and Kuwait have stepped up, replacing much of the lost U.S. volumes.

Starting October 14th, Chinese-owned and/or operated VLGCs calling at U.S. ports will be subject to new fees. This affects 46 live ships (11% of the fleet) and 11 newbuilds (10% of the orderbook). Including ships under Chinese leasing arrangements, the impact is slightly larger. However, the overall effect on the market is expected to be moderate. These ships are now favoring Middle East loadings over U.S., which is increasing competition within the available fleet.

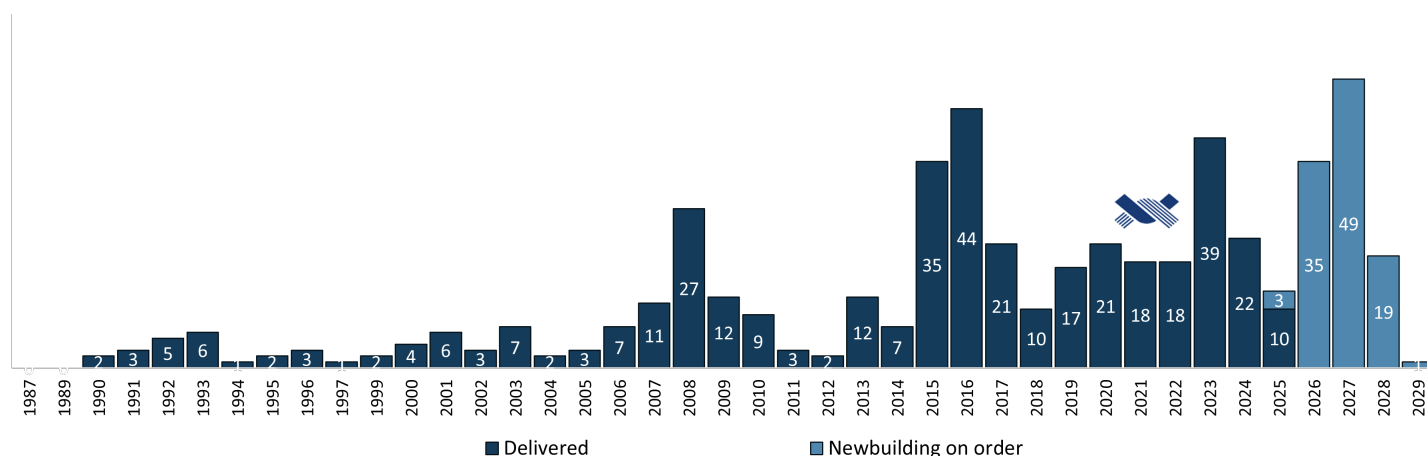


China's LPG demand continues to be primarily driven by the petrochemical sector, with ongoing growth in propane dehydrogenation (PDH) capacity. Although Chinese PDH plants are currently facing negative margins, recent improvements in run rates have been observed, currently at 70%. However, the sector's planned expansions have been heavily reliant on U.S. LPG supply. Should China reduce or phase out sourcing from the U.S., it would have significant implications for both the petrochemical and shipping markets. If propane becomes either too expensive or unavailable, PDH operators may be forced to scale back expansion plans or switch feedstock.

The current tariffs have been imposed by the same U.S. administration that originally introduced them in 2018, suggesting a more resilient and prolonged stance this time around. This could imply that trade disruptions may persist longer than previously anticipated. However, China's dependency on U.S. LPG supply is now five times greater than it was in 2018, potentially giving China greater leverage in negotiations. China currently accounts for over \$10 billion in annual U.S. energy export value, underscoring a strong mutual dependency between the two nations.

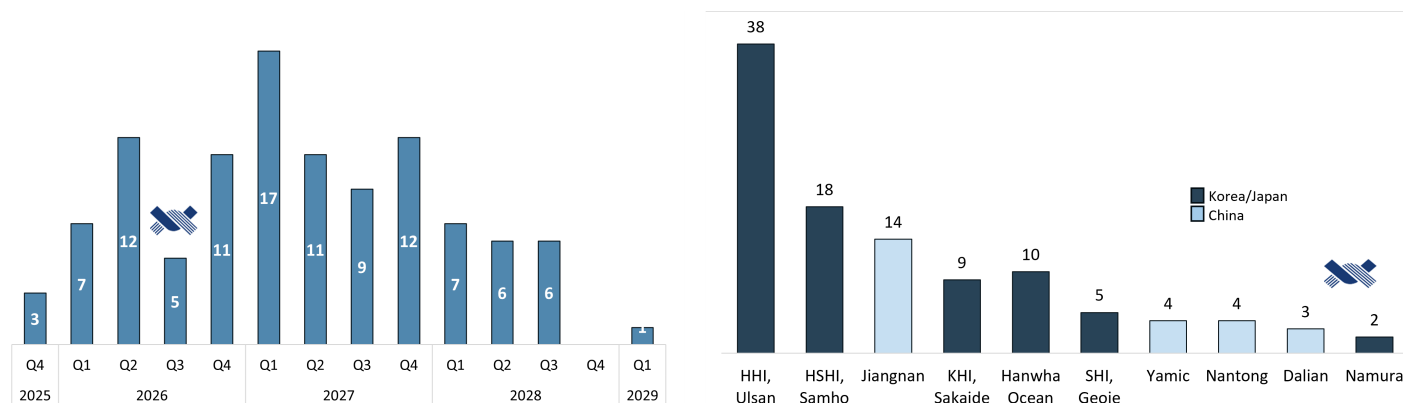
The rapid expansion of China's PDH capacity in recent years means that significant assets are at risk should supply disruptions occur. Conversely, the U.S. could face a surge in stranded LPG exports if China were to halt imports entirely. While alternative buyers exist, an oversupply of product would likely exert downward pressure on U.S. LPG prices. The situation is complex, and while developments are hoped for in the near term, current expectations point to an extension of the existing tariffs. The U.S. administration appears to be focused on other pressing issues, which may delay any resolution on trade policy.

Fleet and Orderbook



As we close in on the last quarter of the year, 10 newbuilds have been delivered, and 3 are remaining, giving a total of 13 newbuilds. The fleet now stands at 410 ships, including floating storages. The orderbook consists of 107 ships, resulting in an orderbook-fleet ratio of 26%, which is slightly above the 20-year average of 21%. Nonetheless, the fleet is set to grow by 19.4% through 2026 and 2027 in total, which could have an accumulated effect by offsetting the fleet balance in to a surplus.

As the Panama Canal is expected to become increasingly congested in the coming years—driven by rising U.S. exports of ethane and LNG, along with the addition of over 100 VLGCs on the water—there is reason to anticipate that a larger share of the fleet will regularly utilize the Cape of Good Hope (COGH) route. This shift will stretch fleet capacity and reduce overall vessel availability. At the same time, new LPG export capacity is coming online in the U.S., requiring additional transportation. These dynamics suggest a potential need for more ships, as longer sailing distances reduce the number of voyages each vessel can complete annually.

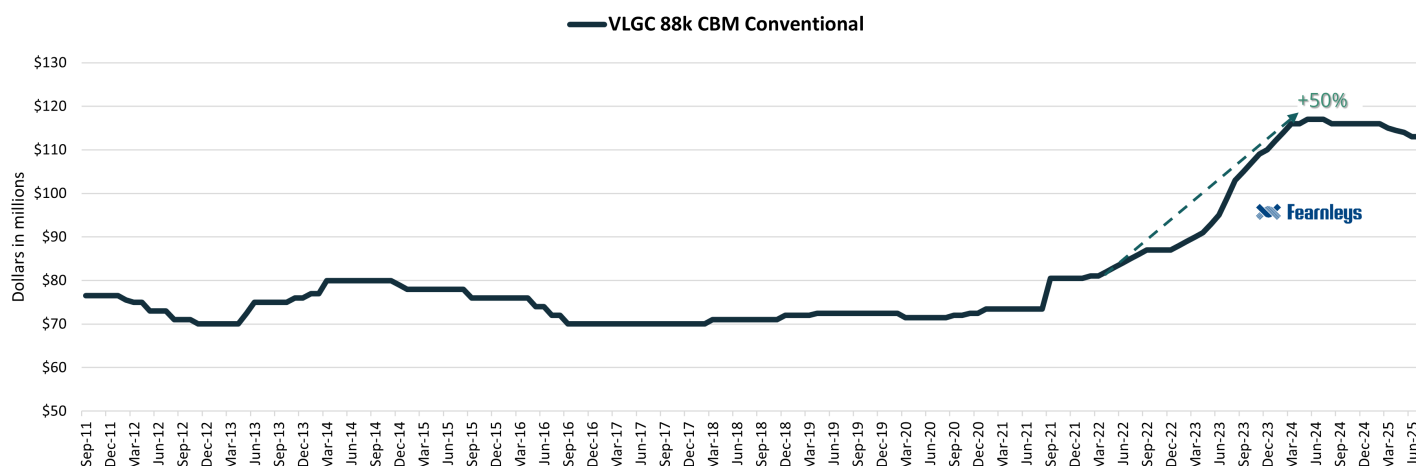


Ordering activity in 2025 has been extremely modest compared to the 2 previous years, as both the current newbuilding prices as well as the size of the orderbook has put a damper on ordering interest.

Hyundai in South Korea remains the largest builder of newbuilds, while one quarter of the newbuilds are being constructed in China. We have not yet seen any difference occurring in asking price for from Japanese/Korean vs. Chinese, and the last price recorded for a VLAC-order was at \$125M in June – Hyundai building for Sahara, with delivery in mid-2028.

On the ammonia side, market interest and activity has remained limited throughout the year. Smaller ship segments are currently under pressure, although the large orderbook for MGCs is expected to accommodate any incremental volumes entering the market. More critically, the infrastructure investments required to support seaborne ammonia trade

via VLACs are not being made. These investments typically involve long lead times, further delaying market readiness. Additionally, the market lacks sufficient project development to generate new ammonia volumes. The first VLAC in the orderbook is scheduled for delivery in June, but as previously noted, these ships are expected to operate as conventional VLGCs for the foreseeable future. We do not differentiate them within the fleet, aside from their technical capability to carry ammonia at 98% capacity, if required.



In line with elevated earning levels, asset values for LPG carriers have risen sharply over the past few years. At the end of 2022, a VLGC was priced at \$80 million—a figure now equivalent to the cost of a 40,000 cbm MGC. Prices increased by 50% to \$120 million as ordering activity surged, reaching record highs for two consecutive years. The peak was recorded in April 2024, when VLGCs were priced at \$127 million. Although ordering activity has slowed in 2025, leading to a softening in newbuilding prices, shipyards continue to maintain relatively high price levels due to long backlogs from other shipping segments. This price softening reflects current market sentiment, which is shaped by a larger orderbook.

The graph above outlines the specifications of a conventional VLGC. Since 2021, nearly all newbuilds have been ordered with LPG dual-fuel capability, reflecting a shift toward cleaner propulsion technologies. From 2023 onward, the VLAC specification has dominated the orderbook. This includes an additional cost of approximately \$2 million for ammonia cargo-handling capability, and a further \$2.5 million for conversion to ammonia dual-fuel engines, assuming the ship's piping and tank systems are pre-configured.

Outlook

In the short term, earnings are expected to remain stable and at similar levels, for the remainder of the year.

Looking ahead to 2026, additional capacity expansions in the U.S., coupled with a projected supply boost from the Middle East, are anticipated to help offset the impact of fleet growth over the next 12 months. Notably, 35 newbuilds are scheduled for delivery next year.

Trade relations between the world's largest LPG importer and exporter remain fluid. However, China is expected to continue diversifying its supply sources to reduce dependency on the U.S.

Transit capacity in the Panama Canal remains constrained under the Long-Term Slot Allocation (LoTSA) system, which the Panama Canal Authority (PCA) has confirmed will continue into 2026. The updated framework includes a reduction in daily transit slots from four to three, and a shortened commitment period from 12 months to two six-month cycles, beginning in January and July. Auctions for these slot packages will commence at the end of October.

This system continues to favor Panamax ships, which benefit from twice as many daily transit slots in the old locks compared to Neo locks. However, with half of the current orderbook consisting of Panamaxes, this dynamic is expected to evolve as fleet composition shifts.



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